

2 Subject content

Qualification aims and objectives

GCSE study in the sciences provides the foundation for understanding the material world. Scientific understanding is changing our lives and is vital to the world's future prosperity. All students should learn essential aspects of the knowledge, methods, processes and uses of science. They should gain appreciation of how the complex and diverse phenomena of the natural world can be described in terms of a small number of key ideas that relate to the sciences and that are both inter-linked and of universal application. These key ideas include:

- the use of conceptual models and theories to make sense of the observed diversity of natural phenomena
- the assumption that every effect has one or more cause
- that change is driven by differences between different objects and systems when they interact
- that many such interactions occur over a distance without direct contact
- that science progresses through a cycle of hypothesis, practical experimentation, observation, theory development and review
- that quantitative analysis is a central element both of many theories and of scientific methods of inquiry.

These key ideas will be assessed through the content of this qualification in the examinations.

These key ideas are relevant in different ways and with different emphases in the three subjects. Examples of their relevance are given for each subject in the separate sections below for Biology, Chemistry and Physics.

The three GCSE Science qualifications enable students to:

- develop scientific knowledge and conceptual understanding through the specific disciplines of Biology, Chemistry and Physics
- develop understanding of the nature, processes and methods of science, through different types of scientific enquiries that help them to answer scientific questions about the world around them
- develop and learn to apply observational, practical, modelling, enquiry and problem-solving skills in the laboratory, in the field and in other learning environments
- develop their ability to evaluate claims based on science through critical analysis of the methodology, evidence and conclusions, both qualitatively and quantitatively.

Students should study the sciences in ways that help them to develop curiosity about the natural world, that give them an insight into how science works and that enable them to appreciate its relevance to their everyday lives. The scope and nature of the study should be broad, coherent, practical and satisfying. It should encourage students to be inspired, motivated and challenged by the subject and its achievements.

The key ideas specific to the Physics content include:

- the use of models, as in the particle model of matter or the wave models of light and of sound
- the concept of cause and effect in explaining such links as those between force and acceleration, or between changes in atomic nuclei and radioactive emissions

- the phenomena of 'action at a distance' and the related concept of the field as the key to analysing electrical, magnetic and gravitational effects
- that differences, for example between pressures or temperatures or electrical potentials, are the drivers of change
- that proportionality, for example between weight and mass of an object or between force and extension in a spring, is an important aspect of many models in science
- that physical laws and models are expressed in mathematical form.

All of these key ideas will be assessed as part of this qualification, through the subject content.

Working scientifically

The GCSE in Physics requires students to develop the skills, knowledge and understanding of working scientifically. Working scientifically will be assessed through examination and the completion of the eight core practicals.

1 Development of scientific thinking

- Understand how scientific methods and theories develop over time.
- Use a variety of models, such as representational, spatial, descriptive, computational and mathematical, to solve problems, make predictions and to develop scientific explanations and an understanding of familiar and unfamiliar facts.
- Appreciate the power and limitations of science, and consider any ethical issues that may arise.
- Explain everyday and technological applications of science; evaluate associated personal, social, economic and environmental implications; and make decisions based on the evaluation of evidence and arguments.
- Evaluate risks both in practical science and the wider societal context, including perception of risk in relation to data and consequences.
- Recognise the importance of peer review of results and of communicating results to a range of audiences.

2 Experimental skills and strategies

- Use scientific theories and explanations to develop hypotheses.
- Plan experiments or devise procedures to make observations, produce or characterise a substance, test hypotheses, check data or explore phenomena.
- Apply a knowledge of a range of techniques, instruments, apparatus and materials to select those appropriate to the experiment.
- Carry out experiments appropriately, having due regard to the correct manipulation of apparatus, the accuracy of measurements and health and safety considerations.
- Recognise when to apply a knowledge of sampling techniques to ensure any samples collected are representative.
- Make and record observations and measurements using a range of apparatus and methods.
- Evaluate methods and suggest possible improvements and further investigations.

3 Analysis and evaluation

Apply the cycle of collecting, presenting and analysing data, including:

- a presenting observations and other data using appropriate methods
- b translating data from one form to another
- c carrying out and representing mathematical and statistical analysis
- d representing distributions of results and making estimations of uncertainty
- e interpreting observations and other data (presented in verbal, diagrammatic, graphical, symbolic or numerical form), including identifying patterns and trends, making inferences and drawing conclusions
- f presenting reasoned explanations, including relating data to hypotheses
- g being objective, evaluating data in terms of accuracy, precision, repeatability and reproducibility and identifying potential sources of random and systematic error
- h communicating the scientific rationale for investigations, methods used, findings and reasoned conclusions through paper-based and electronic reports and presentations using verbal, diagrammatic, graphical, numerical and symbolic forms.

4 Scientific vocabulary, quantities, units, symbols and nomenclature

- a Use scientific vocabulary, terminology and definitions.
- b Recognise the importance of scientific quantities and understand how they are determined.
- c Use SI units (e.g. kg, g, mg; km, m, mm; kJ, J) and IUPAC chemical nomenclature unless inappropriate.
- d Use prefixes and powers of ten for orders of magnitude (e.g. tera, giga, mega, kilo, centi, milli, micro and nano).
- e Interconvert units.
- f Use an appropriate number of significant figures in calculation.

Practical work

The content includes eight mandatory core practicals, indicated as an entire specification point in italics.

Students must carry out all eight of the mandatory core practicals listed below.

Core practical:

- 2.19 *Investigate the relationship between force, mass and acceleration by varying the masses added to trolleys*
- 4.17 *Investigate the suitability of equipment to measure the speed, frequency and wavelength of a wave in a solid and a fluid*
- 5.9 *Investigate refraction in rectangular glass blocks in terms of the interaction of electromagnetic waves with matter*
- 5.19p *Investigate how the nature of a surface affects the amount of thermal energy radiated or absorbed.*
- 10.17 *Construct electrical circuits to:*
 - a investigate the relationship between potential difference, current and resistance for a resistor and a filament lamp*
 - b test series and parallel circuits using resistors and filament lamps*
- 14.3 *Investigate the densities of solid and liquids*
- 14.11 *Investigate the properties of water by determining the specific heat capacity of water and obtaining a temperature-time graph for melting ice*
- 15.6 *Investigate the extension and work done when applying forces to a spring*

Students will need to use their knowledge and understanding of these practical techniques and procedures in the written assessments.

Centres must confirm that each student has completed the eight mandatory core practicals.

Students need to record the work that they have undertaken for the eight mandatory core practicals. The practical record must include the knowledge, skills and understanding they have derived from the practical activities. Centres must complete and submit a Practical Science Statement (see *Appendix 6*) to confirm that all students have completed the eight mandatory core practicals. This must be submitted to Pearson by 15th April in the year that the students will sit their examinations. Any failure by centres to provide this Practical Science Statement will be treated as malpractice and/or maladministration.

Scientific diagrams should be included, where appropriate, to show the set-up and to record the apparatus and procedures used in practical work.

It is important to realise that these core practicals are the minimum number of practicals that should be taken during the course. Suggested additional practicals are given beneath the content at the end of each topic. The eight mandatory core practicals cover all aspects of the apparatus and techniques listed in *Appendix 5: Apparatus and techniques*. This appendix also includes more detailed instructions for each core practical, which must be followed.

Safety is an overriding requirement for all practical work. Centres are responsible for ensuring appropriate safety procedures are followed whenever their students complete practical work.

These core practicals may be reviewed and amended if changes are required to the apparatus and techniques listed by the Department for Education. Pearson may also review and amend the core practicals if necessary. Centres will be told as soon as possible about any changes to core practicals.

Qualification content

The following notation is used in the tables that show the content for this qualification:

- text in **bold** indicates content that is for higher tier only
- entire specification points in italics indicates the core practicals.

Specification statement numbers with a P in them refer to content which is only in the GCSE in Physics and is not found in the GCSE in Combined Science (e.g. 3.18P).

Mathematics

Maths skills that can be assessed in relation to a specification point are referenced in the maths column, next to this specification point. Please see *Appendix 1: Mathematical skills* for full details of each maths skill.

After each topic of content in this specification, there are details relating to the 'Use of mathematics' which contains the Physics specific mathematic skills that are found in each topic of content in the document *Biology, Chemistry and Physics GCSE subject content*, published by the Department for Education (DfE) in June 2014. The reference in brackets after each statement refers to the mathematical skills from *Appendix 1*.

Equations

The required physics equations are listed in *Appendix 2: Equations in Physics*. The first list shows the equations which students are expected to recall for use in the exam papers. These equations may sometimes be given in the exam papers. The equations required for higher tier only are shown in bold text. These equations are also listed in the specification content.

Topics common to Paper 1 and Paper 2

Topic 1 – Key concepts of physics

Students should:		Maths skills
1.1	Recall and use the SI unit for physical quantities, as listed in <i>Appendix 3</i>	
1.2	Recall and use multiples and sub-multiples of units, including giga (G), mega (M), kilo (k), centi (c), milli (m), micro (μ) and nano (n)	3c
1.3	Be able to convert between different units, including hours to seconds	1c
1.4	Use significant figures and standard form where appropriate	1b

Use of mathematics

- Make calculations using ratios and proportional reasoning to convert units and to compute rates (1c, 3c).